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Lens positioning for secondary camera in multi-camera system

Abstract

An example method for camera processing includes detecting a first region of interest for image content captured with a primary camera, detecting a second region of interest for image content captured with a secondary camera, comparing the first region of interest and the second region of interest, determining a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera, determining a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera, determining a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison, and causing the secondary camera to capture images with the secondary camera lens at the determined lens position.

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Background/Summary

(1) This application is a national stage application under 35 U.S.C. § 371 of PCT Application No. PCT/CN2021/090471, filed Apr. 28, 2021. The entire content of PCT Application No. PCT/CN2021/090471 is incorporated herein by reference as if fully set forth below in its entirety and for all applicable purposes.

TECHNICAL FIELD

(2) The disclosure relates to image capture and processing.

BACKGROUND

(3) Image capture devices are incorporated into a wide variety of devices. In this disclosure, an image capture device refers to any device that can capture one or more digital images, including devices that can capture still images and devices that can capture sequences of images to record video. By way of example, image capture devices may comprise stand-alone digital cameras or digital video camcorders, camera-equipped wireless communication device handsets, such as mobile telephones having one or more cameras, cellular or satellite radio telephones, camera-equipped personal digital assistants (PDAs), panels or tablets, gaming devices, computer devices that include cameras, such as so-called “web-cams,” or any devices with digital imaging or video capabilities.

(4) Image capture devices may include multiple cameras (e.g., multiple image sensors and/or multiple lenses) that may be used to support various imaging techniques. An image capture device may capture images with two or more cameras of the multiple cameras and process the images to produce specific image effects, such as depth-of-field effects (e.g., a “bokeh” or “portrait mode” effect).

SUMMARY

(5) In general, this disclosure describes example techniques to determine a position of a lens of a secondary camera for capturing images for image effects such as depth-of-field effects. As described in more detail, a camera processor may determine a lens position for a primary camera and a lens position for a secondary camera based on respective autofocus determinations for the primary camera and the secondary camera. The camera processor may also determine a lens position for the secondary camera based on the determined lens position for the primary camera.

(6) In one or more examples, the camera processor may determine whether the lens position for the secondary camera should be set based on the lens position of the first camera or based on autofocus determination for the secondary camera based on whether one or more criteria are satisfied. Examples of the criteria include whether one or more processors identified a same region of interest (ROI) in images captured with the primary camera and the secondary camera, whether the confidence of the autofocus for the secondary camera is greater than a threshold, and possibly others. In this manner, rather than relying only on the lens position of the primary camera to determine the lens position of the secondary camera, the example techniques described in this disclosure provide for an adaptive manner in which to determine the lens position of the secondary camera, which may result in captured image content that provides better image effects, like depth-of-field effects.

(7) For instance, the example techniques may be considered as a hybrid auto focus technique for the secondary camera auto focus in a multi-camera use case. The image content captured by both primary camera and the secondary camera may have quality sharpness which results in better imaging effects, like for image fusion purpose (e.g., depth-of-field effects) to get more sharpness in the final image content.

(8) In one example, this disclosure describes a device configured for camera processing includes a memory configured to receive images; and one or more processors in communication with the memory, the one or more processors configured to: detect a first region of interest for image content captured with a primary camera; detect a second region of interest for image content captured with a secondary camera; compare the first region of interest and the second region of interest; determine a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; determine a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; determine a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and cause the secondary camera to capture images with the secondary camera lens at the determined lens position.

(9) In another example, this disclosure describes a method for camera processing includes detecting a first region of interest for image content captured with a primary camera; detecting a second region of interest for image content captured with a secondary camera; comparing the first region of interest and the second region of interest; determining a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; determining a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; determining a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and causing the secondary camera to capture images with the secondary camera lens at the determined lens position.

(10) In another example, this disclosure describes a computer-readable storage medium storing instructions thereon that when executed cause one or more processors to: detect a first region of interest for image content captured with a primary camera; detect a second region of interest for image content captured with a secondary camera; compare the first region of interest and the second region of interest; determine a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; determine a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; determine a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and cause the secondary camera to capture images with the secondary camera lens at the determined lens

position.

(11) In another example, this disclosure describes a device for camera processing includes means for detecting a first region of interest for image content captured with a primary camera; means for detecting a second region of interest for image content captured with a secondary camera; means for comparing the first region of interest and the second region of interest; means for determining a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; means for determining a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; means for determining a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and means for causing the secondary camera to capture images with the secondary camera lens at the determined lens position.

(12) This summary is intended to provide an overview of the subject matter described in this disclosure. It is not intended to provide an exclusive or exhaustive explanation of the systems, device, and methods described in detail within the accompanying drawings and description herein. Further details of one or more examples of the disclosed technology are set forth in the accompanying drawings and in the description below. Other features, objects, and advantages of the disclosed technology will be apparent from the description, drawings, and claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a block diagram of a device configured to perform one or more of the example techniques described in this disclosure.

(2) FIG. 2 is a block diagram illustrating example components of the computing device of FIG. 1 in greater detail.

(3) FIG. 3 is a block diagram illustrating example components of an autofocus unit of FIG. 2.

(4) FIG. 4 is a flowchart illustrating an example operation of camera processing in accordance with example techniques of this disclosure.

(5) FIG. 5 is a flowchart illustrating another example operation of camera processing in accordance with example techniques of this disclosure.

(6) Like reference characters denote like elements throughout the description and figures.

DETAILED DESCRIPTION

(7) For image capture, a primary camera (sometimes referred to as a master camera) and a secondary camera (sometimes referred to as a follower or slave camera) work in conjunction for image processing. As one example, for a depth-of-field (e.g., bokeh) effect, the primary camera and secondary camera both capture images, and a camera processor processes the captured images to generate a depth map and produce an image that exhibits a particular depth-of-field. A depth-of-field is the distance between the nearest and farthest object that are in sharp focus. For bokeh effects, the depth-of-field may be fairly shallow.

(8) Bokeh is defined as the effect of a soft out-of-focus background that is achieved when photographing a subject in the foreground (e.g., a subject of portraiture). A bokeh effect can be achieved using post-processing and depth maps on devices with smaller form factors (e.g., mobile phones), even if the camera lenses on such devices do not inherently produce a bokeh effect. For example, a mobile device may include multiple cameras and may capture images with two different cameras (e.g., primary and secondary camera) at the same time or near in time. Because these two cameras are separated by a physical distance, the depth (e.g., distance from the camera sensor) of features in a first image of the two images can be calculated based on a comparison of the features in the first image to the features in the second image. A bokeh effect process may then use the

calculated depth map to blur pixels in the image. For example, pixels and/or features at a depth that matches the focus distance of the image are kept in focus, while pixels that are farther away from the focus distance are blurred. In one example, pixels and/or features that are not at the same depth as the focus distance are all blurred at the same rate. In another example, the farther away a pixel and/or feature is from the focal distance, the more blur that may be applied by the bokeh effect.

(9) For image capture, where a particular depth-of-field may be achieved, such as for bokeh effect, a primary camera is used for image preview and performs autofocus, and a secondary camera is used for depth estimation. The example techniques are described for depth-of-field effects, but may be applicable to other techniques, such as techniques for image fusion (e.g., image content from two or more cameras is fused).

(10) In some examples, the images from both the primary camera and the secondary camera are used to determine the depth estimation. In some techniques, one or more processors determine the lens position for the primary camera lens as part of the autofocus for the primary camera, and the one or more processors determine the lens position for the secondary camera lens based on the position of the primary camera lens. In some cases, determining the lens position for the secondary camera lens based on the lens position for the primary camera can result in incorrect positioning of the secondary camera lens. The incorrect positioning of the secondary camera lens may result in an inaccurate depth map, which may cause depth-of-field effects like bokeh to be of lower quality.

(11) This disclosure describes example techniques in which the one or more processors utilize autofocus to determine the secondary camera lens position. The one or more processors may then set the position of the secondary camera lens to the lens position determined by the autofocus of the secondary camera or to the lens position determined based on the lens position of the primary camera. For example, the secondary camera lens position determined based on the autofocus of the secondary camera may be a first lens position, and the secondary camera lens position determined based on the lens position of the primary camera lens may be a second lens position. The one or more processors may determine whether the secondary lens position should be set to the first lens position or the second lens position based on whether one or more certain criteria are satisfied.

(12) Accordingly, the example techniques may be considered as a hybrid autofocus technique for the secondary camera where the lens position for the secondary camera can be selectively set based on the lens position of the primary camera or based on the lens position determined for the secondary camera. The example techniques may result in sharp image content captured by the primary camera and the secondary camera, which may result in better image effects. For instance, image effects that utilize image fusion (e.g., image content from both the primary and secondary camera) such as Bokeh effect may result in higher quality image content.

(13) As one example, the one or more processors may determine whether the autofocus for both the primary and secondary camera is based on identification of the same regions of interest (e.g., faces from face detection algorithm). If the one or more processors identified different regions of interest in the images captured with the primary and secondary cameras, then the one or more processors may select the second lens position for the secondary camera (e.g., secondary camera lens position determined based on the lens position of the primary camera lens). If the one or more processors identified the same regions of interest, then the one or more processors may determine whether a confidence associated with one or more depth estimation algorithms (e.g., phase detection autofocus (PDAF) or time of flight (TOF) algorithms) determining the depth estimation is greater than a threshold. If the confidence is greater than the threshold, then the one or more processors may select the first lens position for the secondary camera (e.g., secondary camera lens position determined based on the autofocus of the secondary camera). If the confidence is less than the threshold, then the one or more processors may select the second lens position for the secondary camera (e.g., secondary camera lens position determined based on the lens position of the primary camera lens).

(14) The one or more processors may set the lens position for the primary and secondary cameras

based on the respective determined lens positions. The one or more processors may process the captured images, such as to create the bokeh effect, and generate a final image for display.

(15) FIG. 1 is a block diagram of a computing device **10** configured to perform one or more of the example techniques described in this disclosure for lens position for a secondary camera in a multi-camera system. Examples of computing device **10** include a computer (e.g., personal computer, a desktop computer, or a laptop computer), a mobile device such as a tablet computer, a wireless communication device (such as, e.g., a mobile telephone, a cellular telephone, a satellite telephone, and/or a mobile telephone handset), an Internet telephone, a digital camera, a digital video recorder, a handheld device, such as a portable video game device or a personal digital assistant (PDA), a drone device, or any device that may include one or more cameras.

(16) In some examples, computing device **10** may include camera **12A** having lens **13A** and camera **12B** having lens **13B**, camera processor **14**, a central processing unit (CPU) **16**, a graphics processing unit (GPU) **18**, local memory **20** of camera processor **14**, user interface **22**, memory controller **24** that provides access to system memory **30**, and display processor **26** that outputs signals that cause graphical data to be displayed on display **28**. FIG. 1 illustrates two cameras, camera **12A** and camera **12B** (collectively cameras **12**), but there may be additional cameras **12**.

(17) In some examples, cameras **12** need not necessarily be formed on computing device **10**, and may be separate cameras that are coupled to computing device **10** (e.g., via a wired connection). Moreover, although one camera processor **14** is illustrated, in some examples, there may be a plurality of camera processors, like camera processor **14** (e.g., a first camera processor coupled to camera **12A** and a second camera processor coupled to camera **12B**).

(18) Also, there may be more or fewer components on computing device **10** than illustrated. For instance, computing device **10** may include video encoder/decoder for encoding or decoding video, speaker and microphone for communication, antenna and modem for wireless communication, and the like.

(19) Cameras **12** may each include respective arrays of pixel sensors (e.g., pixels) for capturing representations of light, processing circuitry, memory, such as buffer memory or on-chip sensor memory, etc. Lens **13A** and **13B** (collectively lenses **13**) of camera **12A** and camera **12B** may be different types of lens (e.g., each of lenses **13** and image sensor combination having different apertures and/or fields-of-view). Example lenses may include a telephoto lens, a wide angle lens, an ultra-wide angle lens, or other lens types.

(20) Cameras **12** (e.g., via respective lenses **13** and sensors) are configured to capture at least one frame of image data and transfer the at least one frame of the image data to camera processor(s) **14** and/or CPU **16**. In an illustrative example, camera **12A** is configured to transfer a first frame of image data to camera processor(s) **14** and camera **12B** is configured to transfer a second frame of image data to camera processor(s) **14**.

(21) In some examples, camera **12A** may be a primary camera and camera **12B** may be a secondary camera. For instance, camera **12B** may be utilized for depth estimation, such as being a time-of-flight camera used to determine time it takes light outputted by a light emitting diode (LED) of camera **12B** to return back to camera **12B** for determining depth. As another example, during preview mode (e.g., prior to a user capturing image frames that stored long term), display **28** may display image content that is being captured with camera **12A** (e.g., primary camera) and not from camera **12B** (e.g., secondary camera).

(22) For frame capture (e.g., for preview mode, snapshot, or video recording), CPU **16** may be configured to perform autofocus to set the position of lenses **13**. In some examples, CPU **16** may execute “3A” algorithms. Such algorithms may include autofocus (AF), auto exposure control (AEC), and auto white balance (AWB) techniques. In such examples, 3A may represent the functionality of a statistics algorithm processing engine, where CPU **16** may implement and operate such a processing engine. Although the example is described with respect to CPU **16** performing the “3A” algorithms, including performing an autofocus process, the example

techniques are not so limited. Camera processor **14** or other processing circuitry, including processing circuitry of cameras **12** may be configured to execute “3A” algorithms.

(23) In general, the example techniques described in this disclosure may be performed with processing circuitry. Examples of the processing circuitry including processing circuitry of cameras **12**, camera processor **14**, CPU **16**, GPU **18**, display processor **26**, or any combination thereof. For ease of explanation, the example techniques are described with respect to CPU **16**, but the example techniques should not be considered limited to being performed with CPU **16**.

(24) For autofocus, lens **13A** and lens **13B** may be configured to move along an axis to change a distance between lens **13A** and lens **13B** and their respective array of sensors. Lens **13A** and lens **13B** positioned at one end of the axis may result in focus on nearby objects, and lens **13A** and lens **13B** positioned at other end of the axis may result in focus on distant objects.

(25) As described in more detail, this disclosure describes example techniques to determine the lens positions of lens **13A** and lens **13B** of the primary camera **12A** and the secondary camera **12B** that may result in high quality images for display (e.g., correct positioning of lenses **13A** and **13B** for generating an accurate depth map, which may cause depth-of-field effects like bokeh to be of higher quality). For example, CPU **16** may be configured to determine a first lens position for secondary camera lens **13B** for the secondary camera **12B** based on an autofocus process for the secondary camera **12B**. In addition, CPU **16** may be configured to determine a second lens position for the secondary camera lens **13B** of the secondary camera **12B** based on a determined lens position for a primary camera lens **13A** of the primary camera **12A** (e.g., using a mapping function). As described in more detail, CPU **16** may determine a lens position for the secondary camera lens **13B** based on the first lens position or the second lens position based at least in part of certain criteria being satisfied.

(26) In some techniques, CPU **16** may perform an autofocus process on the primary camera (e.g., camera **12A**) to determine a lens position for lens **13A**. Then, rather than performing an autofocus process on the secondary camera (e.g., camera **12B**), CPU **16** may determine the lens position for lens **13B** based on the lens position for lens **13A**. As one example, CPU **16** may apply a mapping function ($F(x)$), where x is the input parameter and the position of lens **13A** and the output of the mapping function is the position of lens **13B**.

(27) In some cases, CPU **16** determining the lens position for lens **13B** based on the lens position for lens **13A** may result in imprecise positioning of lens **13B**. One possible reason for the imprecise positioning of lens **13B** may be because the mapping function (e.g., $F(x)$) for mapping lens position of lens **13B** based on lens position of lens **13A** may not account for differences in manufacturing. For example, the precision of the mapping function may depend on several factors, such as module calibrated anchor data error rate for camera **12B**, degree of temperature rise/burn in for camera **12B**, camera **12B** variance and assembly, and the like.

(28) In accordance with one or more examples described in this disclosure, CPU **16** may be configured to select between a first lens position or a second lens position for secondary camera lens **13B** for secondary camera **12B**. The first lens position may be based on an autofocus process for secondary camera **12B**, and the second lens position may be based on a determined lens position for primary camera lens **13A** for the primary camera **12A**. In this way, in cases where the first lens position is a better lens position (e.g., in terms of producing more accurate depth information for generating quality images with bokeh effect), CPU **16** may be able to set the lens position for lens **13B** to the first position. However, in cases where the second lens position is a better lens position (e.g., in terms of producing more accurate depth information for generating quality images with bokeh effect), CPU **16** may be able to set the lens position for lens **13B** to the second position.

(29) In some examples, camera processor **14** is configured to receive image frames (e.g., pixel data) from cameras **12**, and process the image frames to generate image and/or video content. For example, cameras **12** may be configured to capture individual frames, frame bursts, frame

sequences for generating video content, photo stills captured while recording video, preview frames, or motion photos from before and/or after capture of a still photograph. CPU **16**, GPU **18**, camera processor(s) **14**, or some other circuitry may be configured to process the image and/or video content captured with cameras **12** into images or video for display on display **28**. Image frames may generally refer to frames of data for a still image or frames of video data or combinations thereof, such as with motion photos. Camera processor **14** may receive from cameras **12** pixel data of the image frames in any format. For example, the pixel data may include different color formats, such as RGB, YCbCr, YUV, etc. In any case, camera processor **14** may receive, from cameras **12**, a plurality of frames of image data.

(30) In examples including multiple camera processor(s) **14**, camera processor(s) **14** may share cameras **12**, where each of camera processor(s) **14** may interface with each of camera(s) **12**. In any event, camera processor(s) **14** may initiate capture of a video or image of a scene using a plurality of pixel sensors of camera(s) **12**. In some examples, a video may include a sequence of individual frames. As such, camera processor(s) **14** causes camera(s) **12** to capture the image using the plurality of pixel sensors.

(31) Camera(s) **12** may then output pixel information to camera processor(s) **14** (e.g., pixel values, luma values, color values, charge values, Analog-to-Digital Units (ADU) values, etc.), the pixel information representing the captured image or sequence of captured images. In some examples, camera processor(s) **14** may process monochrome and/or color images to obtain an enhanced color image of a scene. In some examples, camera processor(s) **14** may determine universal blending weight coefficient(s) for different types of pixel blending or may determine different blending weight coefficient(s) for blending different types of pixels that make up a frame of pixels (e.g., a first blending weight coefficient for blending pixels obtained via a monochrome sensor of camera **12A** and pixels obtained via a monochrome sensor of camera **12B**, a second blending weight coefficient for blending pixels obtained via a Bayer sensor of first camera **12A** and pixels obtained via a Bayer sensor of second camera **12B**, etc.).

(32) As described above, CPU **16** may be configured to determine a lens position for the secondary camera lens **13B** based on the first lens position or the second lens position based at least in part on satisfying certain criteria. One example of the criteria to be satisfied is based on a comparison of a region of interest detected in image content captured with primary camera **12A** and a region of interest detected in image content captured with secondary camera **12B**.

(33) As one example, CPU **16** may be configured to detect a first region of interest (ROI) in the image content captured with primary camera **12A** and detect a second ROI in the image content captured with secondary camera **12B**. The ROI may be an object (e.g., face through face detection, pet through pet detection, an object in the foreground, etc.) CPU **16** may be configured to compare the first ROI and the second ROI to determine whether the object in the first ROI is the same as the object in the second ROI (e.g., the same face was detected, the same number of faces were detected, and if multiple faces are detected, the dominant face is the same, etc.). CPU **16** may be configured to determine a lens position for the secondary camera lens **13B** based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest (e.g., based in part on the object in the first ROI being the same as the object in the second ROI).

(34) As another example, the first ROI and the second ROI may be indicative of a location in respective frames captured with the primary camera **12A** and the secondary camera **12B**. For example, the first ROI and the second ROI may be a rectangular region defined by respective coordinates. CPU **16** may be configured to compare a location of the first ROI and a location of the second ROI (e.g., compare lengths and widths of the ROIs, coordinates of the ROIs, etc.). CPU **16** may be configured to determine a lens position for the secondary camera lens **13B** based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest (e.g., based in part on the location of the first

ROI being the same as the location of the second ROI).

(35) In above examples, CPU **16** may determine a lens position for the secondary camera lens **13B** based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest. For instance, if the first region of interest and the second region of interest are the same (e.g., same object in the first ROI and the second ROI or same location of the first ROI and the second ROI), then CPU **16** may determine a lens position of the secondary camera lens **13B** based on the first lens position (e.g., the lens position of secondary camera lens **13B** is the lens position based on an autofocus process for secondary camera **12B**). However, if the first region of interest and the second region of interest are different (e.g., different object in the first ROI and the second ROI or different location of the first ROI and the second ROI), then CPU **16** may determine a lens position of the secondary camera lens **13B** based on the second lens position (e.g., the lens position of secondary camera lens **13B** is based on a lens position of primary camera lens **13A** for primary camera **12A**).

(36) In some examples, even if the first region of interest and the second region of interest are the same, CPU **16** may utilize additional criteria to determine whether a lens position for the secondary camera lens **13B** should be based on the first lens position or the second lens position. As one example, as part of the autofocus, CPU **16** may utilize one or more depth estimation algorithms to determine where to focus. One example of a depth detection algorithm is a phase detection autofocus (PDAF) algorithm. Another example of a depth detection algorithm is a time of flight (TOF) algorithm. The PDAF algorithm and TOF algorithm are described in more detail below.

(37) In one or more examples, the depth detection algorithm may also generate a confidence value indicative of confidence that the autofocus determination is correct. For instance, the depth detection algorithm may indicate how confident the depth detection algorithm is in the determination of where to focus. In some examples, if the confidence value is below a threshold, then CPU **16** may determine that the lens position of the secondary camera lens **13B** should be based on the second lens position (e.g., based on the lens position of the primary camera lens **13A**) even if the first region of interest and the second region of interest are the same.

(38) After CPU **16** determines the lens position for the secondary camera lens **13B** and the lens position for the primary camera lens **13A** (e.g., based on autofocus process for camera **12A**), CPU **16** may cause the secondary camera **12B** to capture images with the secondary camera lens **13B** at the determined lens position, and cause the primary camera **12A** to capture images with the primary camera lens **13A** at the determined lens position. In some examples, CPU **16** and/or camera processor **14** may generate a final image based on images captured by the primary camera **12A** and the secondary camera **12B**. As one example, CPU **16** and/or camera processor **14** may generate a final image having a bokeh effect based on a first image captured with the primary camera **12A** and a second image captured with the secondary camera **12B**.

(39) A bokeh effect may be defined as the effect of a soft out-of-focus background that is achieved when photographing a subject in the foreground (e.g., a portrait). In digital single lens reflex (DSLR) cameras, different bokeh effects may be achieved by using different lenses with different fields-of-view and lens geometries. Bokeh effects are often aesthetically pleasing for portraits when using a telephoto lens at approximately 2× zoom.

(40) A bokeh effect can also be achieved by camera processor **14** by applying post-processing that use depth maps. For example, camera processor **14** may instruct cameras **12** to generate two images at the same time. Because the sensors of cameras **12** are separated by a physical distance, the depth of features of a first image from a primary camera **12A** of the two images can be calculated based on a comparison to those features in a second image generated by a secondary camera **12B**. Camera processor **14** may then apply a bokeh effect on the image produced from camera **12A** by using the calculated depth map to blur features in the image. For example, camera processor **14** may keep in focus features at a depth that matches the focus distance of the generated image, while camera processor **14** may blur features that are farther away from the focus distance. In some examples, the

farther away a pixel is from the focal distance, the more blur that camera processor **14** may apply with the bokeh effect.

(41) Although the various structures of computing device **10** are illustrated as separate in FIG. **1**, the techniques of this disclosure are not so limited, and in some examples the structures may be combined to form a system on chip (SoC). As an example, camera processor **14**, CPU **16**, GPU **18**, and display processor **26** may be formed on a common integrated circuit (IC) chip. In some examples, one or more of camera processor **14**, CPU **16**, GPU **18**, and display processor **26** may be formed on separate IC chips. Various other permutations and combinations are possible, and the techniques of this disclosure should not be considered limited to the example illustrated in FIG. **1**. In an example, CPU **16** may include camera processor **14** such that one or more of camera processor(s) **14** are part of CPU **16**. In such examples, CPU **16** may be configured to perform one or more of the various techniques otherwise ascribed herein to camera processor **14**.

(42) The various structures illustrated in FIG. **1** may be configured to communicate with each other using bus **32**. Bus **32** may be any of a variety of bus structures, such as a third-generation bus (e.g., a HyperTransport bus or an InfiniBand bus), a second-generation bus (e.g., an Advanced Graphics Port bus, a Peripheral Component Interconnect (PCI) Express bus, or an Advanced eXtensible Interface (AXI) bus) or another type of bus or device interconnect. It should be noted that the specific configuration of buses and communication interfaces between the different structures shown in FIG. **1** is merely exemplary, and other configurations of computing devices and/or other image processing systems with the same or different structures may be used to implement the techniques of this disclosure.

(43) In addition, the various components illustrated in FIG. **1** (whether formed on one device or different devices), including cameras **12** and camera processor **14**, may be formed as at least one of fixed-function or programmable circuitry, or a combination of both, such as in one or more microprocessors, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), digital signal processors (DSPs), or other equivalent integrated or discrete logic circuitry. In addition, examples of local memory **20** include one or more volatile or non-volatile memories or storage devices, such as random-access memory (RAM), static RAM (SRAM), dynamic RAM (DRAM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), flash memory, a magnetic data media or an optical storage media.

(44) For example, local memory **20** may provide temporary storage of values produced or utilized by camera processor **14**. In some examples, local memory **20** may also provide memory for other components of computing device **10** such as CPU **16**, GPU **18**, and display processor **26**.

(45) In some examples, memory controller **24** may facilitate the transfer of data going into and out of system memory **30**. For example, memory controller **24** may receive memory read and write commands, and service such commands with respect to memory **30** in order to provide memory services for various components of computing device **10**. In such examples, memory controller **24** may be communicatively coupled to system memory **30**. Although memory controller **24** is illustrated in the example of computing device **10** of FIG. **1** as being a processing circuit that is separate from both CPU **16** and system memory **30**, in some examples, some or all of the functionality of memory controller **24** may be implemented on one or more of CPU **16**, system memory **30**, camera processor **14**, and/or GPU **18**.

(46) System memory **30** may store program modules and/or instructions and/or data that are accessible by camera processor **14**, CPU **16**, and/or GPU **18**. For example, system memory **30** may store user applications (e.g., instructions for a camera application), resulting images from camera processor **14**, etc. System memory **30** may additionally store information for use by and/or generated by other components of computing device **10**. For example, system memory **30** may act as a device memory for camera processor **14**. System memory **30** may include one or more volatile or non-volatile memories or storage devices, such as, for example, RAM, SRAM, DRAM, ROM, EPROM, EEPROM, flash memory, a magnetic data media or an optical storage media. In addition,

system memory **30** may store image data (e.g., frames of video data, encoded video data, sensor-mode settings, zoom settings, 3A parameters, etc.).

(47) In some examples, system memory **30** or local memory **20** may store the image data to on-chip memory, such as in a memory buffer of system memory **30** or local memory **20**. In another example, system memory **30** or local memory **20** may output image data in order to be stored external from the memory of a chip or buffer, such as to a secure digital (SD™) card of a camera device or in some instances, to another internal storage of a camera device. In an illustrative example, system memory **30** or local memory **20** may be embodied as buffer memory on a camera processor **14** chip, GPU **18** chip, or both where a single chip includes both processing circuitries.

(48) In some examples, system memory **30** may include instructions that cause camera processor **14**, CPU **16**, GPU **18**, and/or display processor **26** to perform the functions ascribed to these components in this disclosure. Accordingly, system memory **30** may be a computer-readable storage medium storing instructions stored thereon that, when executed, cause one or more processors (e.g., camera processor **14**, CPU **16**, GPU **18**, and display processor **26**) to perform the various techniques of this disclosure.

(49) In some examples, system memory **30** is a non-transitory storage medium. The term “non-transitory” indicates that the storage medium is not embodied in a carrier wave or a propagated signal. However, the term “non-transitory” should not be interpreted to mean that system memory **30** is non-movable or that its contents are static. As one example, system memory **30** may be removed from computing device **10**, and moved to another device. As another example, memory, substantially similar to system memory **30**, may be inserted into computing device **10**. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM).

(50) In addition, camera processor **14**, CPU **16**, and GPU **18** may store image data, user interface data, etc., in respective buffers that are allocated within system memory **30**. Display processor **26** may retrieve the data from system memory **30** and configure display **28** to display the image represented by the image data, such as via a user interface **22** screen. In some examples, display processor **26** may include a digital-to-analog converter (DAC) that is configured to convert digital values retrieved from system memory **30** into an analog signal consumable by display **28**. In other examples, display processor **26** may pass the digital values directly to display **28** for processing.

(51) CPU **16** may comprise a general-purpose or a special-purpose processor that controls operation of computing device **10**. A user may provide input to computing device **10** to cause CPU **16** to execute one or more software applications. The software applications that execute on CPU **16** may include, for example, a camera application, a graphics editing application, a media player application, a video game application, a graphical user interface application or another program. For example, a camera application may allow the user to control various settings of cameras **12**. The user may provide input to computing device **10** via one or more input devices (not shown) such as a keyboard, a mouse, a microphone, a touch pad or another input device that is coupled to computing device **10** via user interface **22**.

(52) One example software application is a camera application. CPU **16** executes the camera application, and in response, the camera application causes CPU **16** to generate content that display **28** outputs. For instance, display **28** may output information such as light intensity, whether flash is enabled, and other such information. The camera application may also cause CPU **16** to instruct camera processor **14** to process the images output by cameras **12** in a user-defined manner. The user of computing device **10** may interface with display **28** (e.g., via user interface **22**) to configure the manner in which the images are generated (e.g., with zoom settings applied, with or without flash, focus settings, exposure settings, video or still images, and other parameters). For example, CPU **16** may receive via user interface **22** an instruction to apply a depth-of-field effect, such as a bokeh effect (e.g., in a so-called “portrait” mode). The portrait mode may further include a bokeh setting that indicates an amount of blurring to apply.

(53) Display **28** may include a monitor, a television, a projection device, an HDR display, a liquid crystal display (LCD), a plasma display panel, a light emitting diode (LED) array, an organic LED (OLED), electronic paper, a surface-conduction electron-emitted display (SED), a laser television display, a nanocrystal display or another type of display unit. Display **28** may be integrated within computing device **10**. For instance, display **28** may be a screen of a mobile telephone handset, a tablet computer, or a laptop. Alternatively, display **28** may be a stand-alone device coupled to computing device **10** via a wired or wireless communications link. For instance, display **28** may be a computer monitor or flat panel display connected to a personal computer via a cable or wireless link. Display **28** may provide preview frames that a user may view to see what is being stored or what a picture might look like if camera **12A** were to actually take a picture or start recording video.

(54) FIG. **2** is a block diagram illustrating example components of the computing device of FIG. **1** in greater detail. In FIG. **2**, camera **12A** includes sensor **15A**, and camera **12B** includes sensor **15B**. Lens **13A** may direct light towards sensor **15A** that sensor **15A** converts to electrical signals for processing. Similarly, lens **13B** may direct light towards sensor **15B** that sensor **15B** converts to electrical signals for processing.

(55) As illustrated, CPU **16** may be configured to execute ROI detector **34A**, ROI detector **34B**, AF unit **36A**, and AF unit **36B**. Although ROI detector **34A**, ROI detector **34B**, AF unit **36A**, and AF unit **36B** are illustrated as being executed on programmable circuitry of CPU **16**, in one or more examples, ROI detector **34A**, ROI detector **34B**, AF unit **36A**, and AF unit **36B** may be formed as fixed-function circuitry within CPU **16** or external to CPU **16**. In some examples, ROI detector **34A**, ROI detector **34B**, AF unit **36A**, and AF unit **36B** may be formed within camera processor **14**.

(56) ROI detector **34A** may be configured to detect a first region of interest in images captured with camera **12A** (e.g., primary camera), and ROI detector **34B** may be configured to detect a second region of interest in images captured with camera **12B** (e.g., secondary camera). As one example, ROI detector **34A** and ROI detector **34B** may be configured to perform face detection in the images captured by camera **12A** and camera **12B**, respectively. For instance, ROI detector **34A** and ROI detector **34B** may determine a number of faces and location of faces in the images captured by camera **12A** and camera **12B**, respectively.

(57) Faces in the captured images is one example of objects in a region of interest, and should not be considered limiting. Other examples of objects that ROI detector **34A** and ROI detector **34B** may be configured to detect include pets, foreground objects, areas of high contrast, etc. In general, ROI detector **34A** and ROI detector **34B** may be configured to detect objects for detecting the region of interest. The detected region of interests may be located anywhere within the captured images.

(58) AF unit **36A** and AF unit **36B** may be configured to determine a lens position for lens **13A** and lens **13B**, respectively. AF unit **36A** and AF unit **36B** may utilize one or more autofocus techniques, examples of which include, contrast detection auto-focus, phase detection autofocus (PDAF), and time of flight (TOF) autofocus (e.g., laser autofocus). In some examples, PDAF and TOF, or other types of autofocus techniques, may be considered as examples of a depth estimation algorithm because PDAF and TOF determine where to position lens **13A** and lens **13B**, which may be based on how far away the object or area is that is in focus.

(59) For PDAF, AF unit **36A** and AF unit **36B** may utilize phase difference between light captured on different photodiodes of the sensor **15A** via lens **13A** and photodiodes of sensor **15B** via lens **13B**, respectively, to control which area should be focused.

(60) AF unit **36A** may perform PDAF for camera **12A** utilizing the photodiodes of sensor **15A**, without utilizing the photodiodes of sensor **15B**. For example, each photodiode of sensor **15A** may be split in half, and circuitry of camera **12A** or camera processor **14**, as two examples, may measure a phase difference between the two halves of the signal pixel sensor. The photodiodes may be referred to as a 2PD sensor or a dual photodiode sensor. AF unit **36B** may similarly perform PDAF

for camera **12B**. There may be other examples of PDAF techniques as well.

(61) PDAF may be based on comparison of two or more images generated with different asymmetric apertures. When lens **13A** or lens **13B** is not in focus, it is possible to detect disparity caused by the different shape of the asymmetric blur. The size and direction of the disparity is correlated with the defocus amount and direction. AF unit **36A** and AF unit **36B**, may use PDAF implemented at sensor pixel level, using masking layers, dual diodes, or special lens designs. This allows computing device **10** to focus reliably and efficiently in little or no added cost.

(62) In some examples, AF unit **36A** and AF unit **36B** may generate a range of lens positions that AF unit **36A** and AF unit **36B** to evaluate to determine the position of lens **13A** and **13B** using PDAF techniques. For instance, AF unit **36A** and AF unit **36B** may output an initial lens position, referred to as defocus value, where there is zero phase difference between light captured on different photodiodes of the sensors of lens **13A** and lens **13B**. AF unit **36A** and AF unit **36B** may also output a defocus confidence value indicating confidence in the defocus value.

(63) For TOF, a laser or light emitting diode (LED) outputs an artificial light signal that reflects back from objects, and based on the round-trip time of the light, AF unit **36A** and AF unit **36B** can determine the distance of objects. Based on the distance of objects, AF unit **36A** and AF unit **36B** may determine where to autofocus (e.g., focus on nearest object). Similar to PDAF, AF unit **36A** and AF unit **36B** may generate a confidence value indicative of how confident AF unit **36A** and AF unit **36B** are in having correctly determined the nearest object.

(64) In accordance with one or more examples described in this disclosure, AF unit **36A** may determine the lens position for lens **13A** using the example autofocus techniques (e.g., PDAF, TOF, ROI detection from ROI detector **34A**, and the like). AF unit **36B** may, however, determine the lens position for lens **13B** based on a depth estimation algorithm or based on the lens position determined for lens **13A** based on whether certain criteria are satisfied.

(65) For example, AF unit **36B** may determine a first lens position for a secondary camera lens **13B** of the secondary camera **12B** based on an autofocus process for the secondary camera **12B** (e.g., based on PDAF or TOF, or possibly some other technique like contrast detection). AF unit **36B** may determine a second lens position for the secondary camera lens **13B** of the secondary camera **12B** based on a determined lens position for a primary camera lens **13A** of the primary camera **12A**. As one example, AF unit **36B** may utilize a mapping function to determine the second position for the secondary camera lens **13B** of the secondary camera **12B** based on the position of lens **13A** as determined by AF unit **36A**. AF unit **36B** may be configured to determine a lens position for the secondary camera lens **13B** based on the first lens position or the second lens position in response to certain criteria being satisfied.

(66) For example, as described above, ROI detector **34A** may detect a first region of interest for image content captured with a primary camera **12A**, and ROI detector **34B** may detect a second region of interest for image content captured with a secondary camera **12B** (e.g., camera **12B** is utilized for depth estimation). AF unit **36B** may be configured to compare the first region of interest and the second region of interest.

(67) As one example, comparing the first region of interest and the second region of interest may include AF unit **36B** comparing an object in the first region of interest and an object in the second region of interest. In such examples, determining the lens position may include AF unit **36B** determining the lens position for the secondary camera lens **13B** based on the first lens position based at least in part on the object in the first region of interest being the same as the object in the second region of interest. However, if the object in the first region of interest is not the same as the object in the second region of interest, then AF unit **36B** may determine the lens position for the secondary camera lens **13B** based on the second lens position (e.g., based on the mapping function that maps the determined lens position for the primary camera lens **13A** of the primary camera **12A**).

(68) As another example, comparing the first region of interest and the second region of interest

may include AF unit **36B** comparing a location of the first region of interest and a location of the second region of interest. In such examples, determining the lens position may include AF unit **36B** determining the lens position for the secondary camera lens **13B** based on the first lens position based at least in part on the location of the first region of interest being the same as the location of the second region of interest. However, if the location of the first region of interest is not the same as the location of the second region of interest, then AF unit **36B** may determine the lens position for the secondary camera lens **13B** based on the second lens position (e.g., based on the mapping function that maps the determined lens position for the primary camera lens **13A** of the primary camera **12A**).

(69) In the above example, AF unit **36B** determined that the lens position for the secondary camera lens **13B** should be first lens position based on the object in the first region of interest and the second region of interest being the same or based on the location of the first region of interest and the second region of interest being the same. However, in some examples, even if the object in the first region of interest and the second region of interest is the same or the location of the first region of interest and the second region of interest is the same, AF unit **36B** may be configured to determine whether the lens position for the secondary camera lens **13B** should be at the first position or second position based on additional criteria.

(70) As one example, AF unit **36B** may determine a confidence value associated with a depth estimation algorithm for the secondary camera **12B**. The depth estimation algorithm is one of a phase detection autofocus (PDAF) algorithm or a time of flight (TOF) algorithm. For instance, as described above, there may be a confidence value associated with PDAF or TOF. In such examples, determining the lens position for the secondary camera lens **13B** may include determining the lens position for the secondary camera lens **13B** based on the first lens position based on the comparison and the confidence value being greater than a threshold value.

(71) That is, if the comparison of the first region of interest and the second region of interest indicates that the objects are the same (e.g., including that the same number of objects are detected) and/or the locations are the same, and the confidence value associated with the depth estimation algorithm is greater than the threshold value, then AF unit **36B** may determine the lens position for the secondary camera lens **13B** based on the first lens position (e.g., where the first lens position is based on autofocus process for the secondary camera **12B**). However, if the comparison of the first region of interest and the second region of interest indicates that the objects are the same (e.g., including that the same number of objects are detected) and/or the locations are the same, but the confidence value associated with the depth estimation algorithm is less than or equal to the threshold value, then AF unit **36B** may determine the lens position for the secondary camera lens **13B** based on the second lens position (e.g., where the second lens position is based on a determined lens position for primary camera lens **13A** of the primary camera **12A**).

(72) In some examples, AF unit **36B** may utilize both the PDAF and TOF algorithms. For example, AF unit **36B** may determine a first confidence value associated with a first depth estimation algorithm (e.g., PDAF) for the secondary camera **12B** and a second confidence value associated with a second depth estimation algorithm (e.g., TOF) for the secondary camera **12B**. In this example, determining the lens position for the secondary camera lens **13B** may include AF unit **36B** determining the lens position for the secondary camera lens **13B** based on the first lens position based on the comparison, the first confidence value being less than a first threshold value, and the second confidence value being greater than a second threshold value.

(73) That is, if the comparison of the first region of interest and the second region of interest indicates that the objects are the same (e.g., including that the same number of objects are detected) and/or the locations are the same, and the first confidence value associated with the first depth estimation algorithm (e.g., PDAF) is less than the first threshold value, then AF unit **36B** may determine whether the second confidence value associated with the second depth estimation algorithm (e.g., TOF) is greater than the second threshold value. If the second confidence value

associated with the second depth estimation algorithm is greater than the second threshold value, then AF unit **36B** may determine the lens position for the secondary camera lens **13B** based on the first lens position (e.g., based on an autofocus process for secondary camera **12B**). If the second confidence value associated with the second depth estimation algorithm is less than or equal to the second threshold value, then AF unit **36B** may determine the lens position for the secondary camera lens **13B** based on the second lens position (e.g., based on a determined lens position for primary camera lens **13A** of primary camera **12A**).

(74) FIG. **3** is a block diagram illustrating example components of an autofocus unit of FIG. **2**. For example, FIG. **3** illustrates AF unit **36B** in further detail. As illustrated, AF unit **36B** includes position mapping unit **40**, PDAF unit **42**, and TOF unit **44**. PDAF unit **42** and/or TOF unit **44** may be configured to determine a first lens position for secondary camera lens **13B** of secondary camera **12B** based on an autofocus process (e.g., PDAF or TOF, respectively) for secondary camera **12B**. In addition, PDAF unit **42** may determine a first confidence value indicative of the confidence of the lens position as determined based on PDAF, and TOF unit **44** may determine a second confidence value indicative of the confidence of objects as determined based on TOF. Position mapping unit **40** may be configured to determine a second lens position for secondary camera lens **13B** of secondary camera **12B** based on a determined lens position for a primary camera lens **13A** of the primary camera **12A** (e.g., as determined by AF unit **36A**).

(75) Aggregate unit **46** may determine a lens position for the secondary camera lens **13B** based on the first lens position or the second lens position. For instance, aggregate unit **46** may receive the first region of interest for image content captured with a primary camera **12A** detected by ROI detector **34A**, receive the second region of interest for image content captured with a secondary camera **12B**, receive the first confidence value associated with PDAF unit **42**, and receive a second confidence value associated with TOF unit **44**. Aggregate unit **46** may utilize the received information to determine a lens position for the secondary camera lens **13B** based on the first lens position or the second lens position.

(76) To ease with understanding, the example of FIG. **3** is described with respect to FIG. **4**, which illustrates a method for camera processing. For instance, FIG. **4** is a flowchart illustrating an example operation of camera processing in accordance with example techniques of this disclosure.

(77) Aggregate unit **46** may determine whether a first criteria is satisfied (**50**). An example of the first criteria may be whether there is only one object to detect (e.g., only one face), and both the first region of interest and the second region of interest are the same object or the same location. For example, aggregate unit **46** may compare the first region of interest and the second region of interest to determine whether an object in the first region of interest and an object in the second region of interest is the same or to determine whether a location of the first region of interest and a location of the second region of interest is the same.

(78) If the first criteria is not satisfied (NO of **50**), then aggregate unit **46** may determine whether a second criteria is satisfied (**52**). An example of the second criteria may be whether there are two or more objects to detect (e.g., two or more faces), and both the first region of interest and the second region of interest are the same dominant object (e.g., dominant face). The second criteria and the first criteria may be the same except that in the second criteria, there are multiple objects. For the first and/or second criteria to be satisfied, the first region of interest and the second region of interest are the same object or the same location.

(79) If the first criteria is satisfied (YES of **50**) or the second criteria is satisfied (YES of **52**), aggregate unit **46** may determine whether the first confidence value (e.g., confidence value associated with PDAF) is greater than a first threshold value (**54**). If aggregate unit **46** determines that the first confidence value is greater than the first threshold value (YES of **54**), then aggregate unit **46** may determine a lens position for the secondary camera lens **13B** based on the first lens position (e.g., based on the PDAF autofocus process for the secondary camera **12B**) (**56**).

(80) If aggregate unit **46** determines that the first confidence value is less than or equal to the first

threshold value (NO of 54), then aggregate unit 46 may determine whether the second confidence value (e.g., confidence value associated with TOF) is greater than a second threshold value (58). If aggregate unit 46 determines that the second confidence value is greater than the second threshold value (YES of 58), then aggregate unit 46 may determine a lens position for the secondary camera lens 13B based on the first lens position (e.g., based on the TOF autofocus process for the secondary camera 12B) (60).

(81) If aggregate unit 46 determines that the second criteria is not satisfied (NO of 52) (e.g., aggregate unit 46 determines that an object in the first region of interest is different than an object in the second region of interest) or if the second confidence value is less than second threshold value (NO of 58), the aggregate unit 46 may determine a lens position for the secondary camera lens 13B based on the second lens position (e.g., based on a determined lens position for a primary camera lens 13A of the primary camera 12A) (62). For example, aggregate unit 46 may determine the lens position for secondary camera lens 13B to be the output from position mapping unit 40. Position mapping unit 40 may be configured to apply the mapping function ($F(x)$) to determine the second lens position based on a determined lens position for a primary camera lens 13A of the primary camera 12A.

(82) In the example of FIG. 4, both the confidence value associated with PDAF and the confidence value associated with TOF are utilized. However, the example techniques are not so limited. For instance, based on the first criteria being satisfied (YES of 50) or the second criteria being satisfied (YES of 52), aggregate unit 46 may determine whether a confidence value (e.g., confidence value associated to PDAF or confidence value associated with TOF) is greater than a threshold value. Based on the confidence value being greater than the threshold value, aggregate unit 46 may determine the lens position for secondary camera lens 13B based on the first lens position (e.g., lens position determined by either PDAF unit 42 or TOF unit 44).

(83) FIG. 5 is a flowchart illustrating another example operation of camera processing in accordance with example techniques of this disclosure. For ease, the method for camera processing example of FIG. 5 is described with respect to one or more processors in communication with memory configured to receive images. Examples of the one or more processors include camera processor 14, CPU 16, GPU 18, processing circuitry of cameras 12A and 12B, display processor 26, or other processing circuitry of computing device 10. Examples of the memory include memory 30, local memory 20, or other memory for computing device 10.

(84) The one or more processors may detect a first region of interest for image content captured with a primary camera 12A (70). For example, ROI detector 34A may detect the first region of interest for image content captured with primary camera 12A. The one or more processors may detect a second region of interest for image content captured with a secondary camera 12B (72). For example, ROI detector 34B may detect the second region of interest for image content captured with secondary camera 12B.

(85) The one or more processors may compare the first region of interest and the second region of interest (74). For example, AF unit 36B (e.g., via aggregate unit 46) may compare an object in the first region of interest and an object in the second region of interest (e.g., to determine if the objects are the same). As another example, AF unit 36B (e.g., via aggregate unit 46) may compare a location of the first region of interest and a location of the second region of interest (e.g., to determine if the locations are the same). In some examples, aggregate unit 46 may also determine whether the same number of objects are determined (e.g., only one face or a plurality of faces).

(86) The one or more processors may determine a first lens position for a secondary camera lens 13B of the secondary camera 12B based on an autofocus process for the secondary camera 12B (76). For example, PDAF unit 42 may determine a lens position for a secondary camera lens 13B based on the PDAF autofocus process for secondary camera 12B, and determine a confidence value associated with PDAF (e.g., first confidence value). TOF unit 44 may determine a lens position for a secondary camera lens 13B based on the TOF autofocus process for secondary camera 12B, and

determine a confidence value associated with TOF (e.g., second confidence value).

(87) The one or more processors may determine a second lens position for the secondary camera lens **13B** of the secondary camera **12B** based on a determined lens position for a primary camera lens **13A** of the primary camera **12A** (**78**). For example, position mapping unit **40** may determine the second lens position based on the lens position for primary camera lens **13A** as determined by AF unit **36A**.

(88) The one or more processors may determine a lens position for the secondary camera lens **13B** based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest (**80**). For example, aggregate unit **46** may determine the lens position for the secondary camera lens **13B** based on the first lens position based at least in part on the object in the first region of interest being the same as the object in the second region of interest. For instance, aggregate unit **46** may set the lens position for the secondary camera lens **13B** equal to the first lens position.

(89) However, aggregate unit **46** may determine the lens position for the secondary camera lens **13B** based on the second lens position based at least in part on the comparison indicating that an object in the first region of interest is different than an object in the second region of interest. For instance, aggregate unit **46** may set the lens position for the secondary camera lens **13B** equal to the second lens position.

(90) In some examples, for determining the lens position for the secondary camera lens **13B**, aggregate unit **46** may determine the lens position for the secondary camera lens **13B** based on the first lens position based on the comparison and the confidence value (e.g., the confidence value associated with PDAF or TOF) being greater than a threshold value. For example, if the comparison indicates that the first region of interest and the second region of interest are the same (e.g., same object, same location, etc.), then aggregate unit **46** may determine whether the confidence value is greater than a threshold value. If the confidence value is greater than the threshold value, then aggregate unit **46** may determine the lens position for the secondary camera lens **13B** based on the first lens position (e.g., lens position for secondary camera lens **13B** equals first lens position). If the confidence value is less than or equal to the threshold value, then aggregate unit **46** may determine the lens position for the secondary camera lens **13B** based on the second lens position (e.g., lens position for secondary camera lens **13B** equals second lens position).

(91) In some examples, for determining the lens position for the secondary camera lens **13B**, aggregate unit **46** may determine the lens position for the secondary camera lens based on the first lens position based on the comparison, the first confidence value (e.g., confidence value associated with PDAF) being less than a first threshold value, and the second confidence value (e.g., confidence value associated with TOF) being greater than a second threshold value. For example, if the comparison indicates that the first region of interest and the second region of interest are the same (e.g., same object, same location, etc.), then aggregate unit **46** may determine whether the first confidence value is greater than a first threshold value.

(92) If the first confidence value is less than or equal to the first threshold value, then aggregate unit **46** may determine whether the second confidence value is greater than a second threshold value. If the second confidence value is greater than the second threshold value, then aggregate unit **46** may determine the lens position for the secondary camera lens **13B** based on the first lens position (e.g., lens position for secondary camera lens **13B** equals first lens position). If the second confidence value is less than or equal to the second threshold value, then aggregate unit **46** may determine the lens position for the secondary camera lens **13B** based on the second lens position (e.g., lens position for secondary camera lens **13B** equals second lens position).

(93) The one or more processors may cause the secondary camera **12B** to capture images with the secondary camera lens **13B** at the determined lens position (**82**). For example, the one or more processors may cause primary camera **12A** to capture images with the primary camera lens **13A** at

its determined lens position. The one or more processors may perform image processing to generate image content for display. As one example, the one or more processors may generate a final image having a bokeh effect based on a first image captured with the primary camera and a second image captured with the secondary camera.

(94) The following describe one or more example techniques described in this disclosure. The example techniques may be used together or separately.

(95) Clause 1: A device configured for camera processing includes a memory configured to receive images; and one or more processors in communication with the memory, the one or more processors configured to: detect a first region of interest for image content captured with a primary camera; detect a second region of interest for image content captured with a secondary camera; compare the first region of interest and the second region of interest; determine a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; determine a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; determine a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and cause the secondary camera to capture images with the secondary camera lens at the determined lens position.

(96) Clause 2: The device of clause 1, wherein the one or more processors are configured to generate a final image having a bokeh effect based on a first image captured with the primary camera and a second image captured with the secondary camera.

(97) Clause 3: The device of any of clauses 1 and 2, wherein comparing the first region of interest and the second region of interest comprises comparing an object in the first region of interest and an object in the second region of interest, and wherein determining the lens position comprises determining the lens position for the secondary camera lens based on the first lens position based at least in part on the object in the first region of interest being the same as the object in the second region of interest.

(98) Clause 4: The device of any of clauses 1 through 3, wherein comparing the first region of interest and the second region of interest comprises comparing a location of the first region of interest and a location of the second region of interest, and wherein determining the lens position comprises determining the lens position for the secondary camera lens based on the first lens position based at least in part on the location of the first region of interest being the same as the location of the second region of interest.

(99) Clause 5: The device of any of clauses 1 through 4, wherein the one or more processors are configured to: determine a confidence value associated with a depth estimation algorithm for the secondary camera, wherein determining the lens position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the first lens position based on the comparison and the confidence value being greater than a threshold value.

(100) Clause 6: The device of clause 5, wherein the depth estimation algorithm is one of a phase detection autofocus (PDAF) algorithm or a time of flight (TOF) algorithm.

(101) Clause 7: The device of any of clauses 1 through 6, wherein the one or more processors are configured to: determine a first confidence value associated with a first depth estimation algorithm for the secondary camera and a second confidence value associated with a second depth estimation algorithm for the secondary camera, wherein determining the lens position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the first lens position based on the comparison, the first confidence value being less than a first threshold value, and the second confidence value being greater than a second threshold value.

(102) Clause 8: The device of clause 7, wherein the first depth estimation algorithm is a phase detection autofocus (PDAF) algorithm, and the second depth estimation algorithm is a time of flight (TOF) algorithm.

(103) Clause 9: The device of any of clauses 1 through 8, wherein determining the lens position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the second lens position based on the comparison indicating that an object in the first region of interest being different than an object in the second region of interest.

(104) Clause 10: The device of any of clauses 1 through 9, wherein the device comprises the primary camera and the secondary camera.

(105) Clause 11: A method for camera processing includes detecting a first region of interest for image content captured with a primary camera; detecting a second region of interest for image content captured with a secondary camera; comparing the first region of interest and the second region of interest; determining a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; determining a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; determining a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and causing the secondary camera to capture images with the secondary camera lens at the determined lens position.

(106) Clause 12: The method of clause 11, further includes generating a final image having a bokeh effect based on a first image captured with the primary camera and a second image captured with the secondary camera.

(107) Clause 13: The method of any of clauses 11 and 12, wherein comparing the first region of interest and the second region of interest comprises comparing an object in the first region of interest and an object in the second region of interest, and wherein determining the lens position comprises determining the lens position for the secondary camera lens based on the first lens position based at least in part on the object in the first region of interest being the same as the object in the second region of interest.

(108) Clause 14: The method of any of clauses 11 through 13, wherein comparing the first region of interest and the second region of interest comprises comparing a location of the first region of interest and a location of the second region of interest, and wherein determining the lens position comprises determining the lens position for the secondary camera lens based on the first lens position based at least in part on the location of the first region of interest being the same as the location of the second region of interest.

(109) Clause 15: The method of any of clauses 11 through 14, further includes determining a confidence value associated with a depth estimation algorithm for the secondary camera, wherein determining the lens position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the first lens position based on the comparison and the confidence value being greater than a threshold value.

(110) Clause 16: The method of clause 15, wherein the depth estimation algorithm is one of a phase detection autofocus (PDAF) algorithm or a time of flight (TOF) algorithm.

(111) Clause 17: The method of any of clauses 11 through 16, further includes determining a first confidence value associated with a first depth estimation algorithm for the secondary camera and a second confidence value associated with a second depth estimation algorithm for the secondary camera, wherein determining the lens position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the first lens position based on the comparison, the first confidence value being less than a first threshold value, and the second confidence value being greater than a second threshold value.

(112) Clause 18: The method of clause 17, wherein the first depth estimation algorithm is a phase detection autofocus (PDAF) algorithm, and the second depth estimation algorithm is a time of flight (TOF) algorithm.

(113) Clause 19: The method of any of clauses 11 through 18, wherein determining the lens

position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the second lens position based on the comparison indicating that an object in the first region of interest being different than an object in the second region of interest.

(114) Clause 20: A computer-readable storage medium storing instructions thereon that when executed cause one or more processors to: detect a first region of interest for image content captured with a primary camera; detect a second region of interest for image content captured with a secondary camera; compare the first region of interest and the second region of interest; determine a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; determine a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; determine a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and cause the secondary camera to capture images with the secondary camera lens at the determined lens position.

(115) Clause 21: The computer-readable storage medium of clause 20, further includes generate a final image having a bokeh effect based on a first image captured with the primary camera and a second image captured with the secondary camera.

(116) Clause 22: The computer-readable storage medium of any of clauses 20 and 21, wherein the instructions that cause the one or more processors to compare the first region of interest and the second region of interest comprise instructions that cause the one or more processors to compare an object in the first region of interest and an object in the second region of interest, and wherein the instructions that cause the one or more processors to determine the lens position comprise instructions that cause the one or more processors to determine the lens position for the secondary camera lens based on the first lens position based at least in part on the object in the first region of interest being the same as the object in the second region of interest.

(117) Clause 23: The computer-readable storage medium of any of clauses 20 through 22, wherein the instructions that cause the one or more processors to compare the first region of interest and the second region of interest comprise instructions that cause the one or more processors to compare a location of the first region of interest and a location of the second region of interest, and wherein the instructions that cause the one or more processors to determine the lens position comprise instructions that cause the one or more processors to determine the lens position for the secondary camera lens based on the first lens position based at least in part on the location of the first region of interest being the same as the location of the second region of interest.

(118) Clause 24: The computer-readable storage medium of any of clauses 20 through 23, further includes determine a confidence value associated with a depth estimation algorithm for the secondary camera, wherein the instructions that cause the one or more processors to determine the lens position for the secondary camera lens comprise instructions that cause the one or more processors to determine the lens position for the secondary camera lens based on the first lens position based on the comparison and the confidence value being greater than a threshold value.

(119) Clause 25: The computer-readable storage medium of any of clauses 20 through 24, further includes determine a first confidence value associated with a first depth estimation algorithm for the secondary camera and a second confidence value associated with a second depth estimation algorithm for the secondary camera, wherein the instructions that cause the one or more processors to determine the lens position for the secondary camera lens comprise instructions that cause the one or more processors to determine the lens position for the secondary camera lens based on the first lens position based on the comparison, the first confidence value being less than a first threshold value, and the second confidence value being greater than a second threshold value.

(120) Clause 26: A device for camera processing includes means for detecting a first region of interest for image content captured with a primary camera; means for detecting a second region of interest for image content captured with a secondary camera; means for comparing the first region

of interest and the second region of interest; means for determining a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; means for determining a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; means for determining a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and means for causing the secondary camera to capture images with the secondary camera lens at the determined lens position.

(121) Clause 27: The device of clause 26, further includes means for generating a final image having a bokeh effect based on a first image captured with the primary camera and a second image captured with the secondary camera.

(122) Clause 28: The device of any of clauses 26 and 27, wherein the means for comparing the first region of interest and the second region of interest comprises means for comparing an object in the first region of interest and an object in the second region of interest, and wherein the means for determining the lens position comprises means for determining the lens position for the secondary camera lens based on the first lens position based at least in part on the object in the first region of interest being the same as the object in the second region of interest.

(123) Clause 29: The device of any of clauses 26 through 28, wherein the means for comparing the first region of interest and the second region of interest comprises means for comparing a location of the first region of interest and a location of the second region of interest, and wherein the means for determining the lens position comprises means for determining the lens position for the secondary camera lens based on the first lens position based at least in part on the location of the first region of interest being the same as the location of the second region of interest.

(124) Clause 30: The device of any of clauses 26 through 29, further includes means for determining a confidence value associated with a depth estimation algorithm for the secondary camera, wherein the means for determining the lens position for the secondary camera lens comprises means for determining the lens position for the secondary camera lens based on the first lens position based on the comparison and the confidence value being greater than a threshold value.

(125) In one or more examples, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over, as one or more instructions or code, a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include computer-readable storage media, which corresponds to a tangible medium such as data storage media. In this manner, computer-readable media generally may correspond to tangible computer-readable storage media which is non-transitory. Data storage media may be any available media that can be accessed by one or more computers or one or more processors to retrieve instructions, code and/or data structures for implementation of the techniques described in this disclosure. A computer program product may include a computer-readable medium.

(126) By way of example, and not limitation, such computer-readable storage media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage, or other magnetic storage devices, flash memory, cache memory, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer. It should be understood that computer-readable storage media and data storage media do not include carrier waves, signals, or other transient media, but are instead directed to non-transient, tangible storage media. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc, where discs usually reproduce data magnetically, while discs reproduce data optically with lasers.

Combinations of the above should also be included within the scope of computer-readable media.

(127) Instructions may be executed by one or more processors, such as one or more digital signal

processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), field programmable logic arrays (FPGAs), or other equivalent integrated or discrete logic circuitry. Accordingly, the term “processor,” as used herein may refer to any of the foregoing structure or any other structure suitable for implementation of the techniques described herein. Also, the techniques could be fully implemented in one or more circuits or logic elements.

(128) The techniques of this disclosure may be implemented in a wide variety of devices or apparatuses, including a wireless handset, an integrated circuit (IC) or a set of ICs (e.g., a chip set). Various components, modules, or units are described in this disclosure to emphasize functional aspects of devices configured to perform the disclosed techniques, but do not necessarily require realization by different hardware units.

(129) Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. A device configured for camera processing, the device comprising: a memory configured to receive images; and one or more processors in communication with the memory, the one or more processors configured to: detect a first region of interest for image content captured with a primary camera; detect a second region of interest for image content captured with a secondary camera; compare the first region of interest and the second region of interest; determine a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; determine a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; determine a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and cause the secondary camera to capture images with the secondary camera lens at the determined lens position.
2. The device of claim 1, wherein the one or more processors are configured to generate a final image having a bokeh effect based on a first image captured with the primary camera and a second image captured with the secondary camera.
3. The device of claim 1, wherein comparing the first region of interest and the second region of interest comprises comparing an object in the first region of interest and an object in the second region of interest, and wherein determining the lens position comprises determining the lens position for the secondary camera lens based on the first lens position based at least in part on the object in the first region of interest being the same as the object in the second region of interest.
4. The device of claim 1, wherein comparing the first region of interest and the second region of interest comprises comparing a location of the first region of interest and a location of the second region of interest, and wherein determining the lens position comprises determining the lens position for the secondary camera lens based on the first lens position based at least in part on the location of the first region of interest being the same as the location of the second region of interest.
5. The device of claim 1, wherein the one or more processors are configured to: determine a confidence value associated with a depth estimation algorithm for the secondary camera, wherein determining the lens position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the first lens position based on the comparison and the confidence value being greater than a threshold value.
6. The device of claim 5, wherein the depth estimation algorithm is one of a phase detection autofocus (PDAF) algorithm or a time of flight (TOF) algorithm.
7. The device of claim 1, wherein the one or more processors are configured to: determine a first confidence value associated with a first depth estimation algorithm for the secondary camera and a second confidence value associated with a second depth estimation algorithm for the secondary

camera, wherein determining the lens position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the first lens position based on the comparison, the first confidence value being less than a first threshold value, and the second confidence value being greater than a second threshold value.

8. The device of claim 7, wherein the first depth estimation algorithm is a phase detection autofocus (PDAF) algorithm, and the second depth estimation algorithm is a time of flight (TOF) algorithm.

9. The device of claim 1, wherein determining the lens position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the second lens position based on the comparison indicating that an object in the first region of interest being different than an object in the second region of interest.

10. The device of claim 1, wherein the device comprises the primary camera and the secondary camera.

11. A method for camera processing, the method comprising: detecting a first region of interest for image content captured with a primary camera; detecting a second region of interest for image content captured with a secondary camera; comparing the first region of interest and the second region of interest; determining a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; determining a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; determining a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and causing the secondary camera to capture images with the secondary camera lens at the determined lens position.

12. The method of claim 11, further comprising: generating a final image having a bokeh effect based on a first image captured with the primary camera and a second image captured with the secondary camera.

13. The method of claim 11, wherein comparing the first region of interest and the second region of interest comprises comparing an object in the first region of interest and an object in the second region of interest, and wherein determining the lens position comprises determining the lens position for the secondary camera lens based on the first lens position based at least in part on the object in the first region of interest being the same as the object in the second region of interest.

14. The method of claim 11, wherein comparing the first region of interest and the second region of interest comprises comparing a location of the first region of interest and a location of the second region of interest, and wherein determining the lens position comprises determining the lens position for the secondary camera lens based on the first lens position based at least in part on the location of the first region of interest being the same as the location of the second region of interest.

15. The method of claim 11, further comprising: determining a confidence value associated with a depth estimation algorithm for the secondary camera, wherein determining the lens position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the first lens position based on the comparison and the confidence value being greater than a threshold value.

16. The method of claim 15, wherein the depth estimation algorithm is one of a phase detection autofocus (PDAF) algorithm or a time of flight (TOF) algorithm.

17. The method of claim 11, further comprising: determining a first confidence value associated with a first depth estimation algorithm for the secondary camera and a second confidence value associated with a second depth estimation algorithm for the secondary camera, wherein determining the lens position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the first lens position based on the comparison, the first confidence value being less than a first threshold value, and the second confidence value being

greater than a second threshold value.

18. The method of claim 17, wherein the first depth estimation algorithm is a phase detection autofocus (PDAF) algorithm, and the second depth estimation algorithm is a time of flight (TOF) algorithm.

19. The method of claim 11, wherein determining the lens position for the secondary camera lens comprises determining the lens position for the secondary camera lens based on the second lens position based on the comparison indicating that an object in the first region of interest being different than an object in the second region of interest.

20. A computer-readable storage medium storing instructions thereon that when executed cause one or more processors to: detect a first region of interest for image content captured with a primary camera; detect a second region of interest for image content captured with a secondary camera; compare the first region of interest and the second region of interest; determine a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; determine a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; determine a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and cause the secondary camera to capture images with the secondary camera lens at the determined lens position.

21. The computer-readable storage medium of claim 20, further comprising instructions that cause the one or more processors to: generate a final image having a bokeh effect based on a first image captured with the primary camera and a second image captured with the secondary camera.

22. The computer-readable storage medium of claim 20, wherein the instructions that cause the one or more processors to compare the first region of interest and the second region of interest comprise instructions that cause the one or more processors to compare an object in the first region of interest and an object in the second region of interest, and wherein the instructions that cause the one or more processors to determine the lens position comprise instructions that cause the one or more processors to determine the lens position for the secondary camera lens based on the first lens position based at least in part on the object in the first region of interest being the same as the object in the second region of interest.

23. The computer-readable storage medium of claim 20, wherein the instructions that cause the one or more processors to compare the first region of interest and the second region of interest comprise instructions that cause the one or more processors to compare a location of the first region of interest and a location of the second region of interest, and wherein the instructions that cause the one or more processors to determine the lens position comprise instructions that cause the one or more processors to determine the lens position for the secondary camera lens based on the first lens position based at least in part on the location of the first region of interest being the same as the location of the second region of interest.

24. The computer-readable storage medium of claim 20, further comprising instructions that cause the one or more processors to: determine a confidence value associated with a depth estimation algorithm for the secondary camera, wherein the instructions that cause the one or more processors to determine the lens position for the secondary camera lens comprise instructions that cause the one or more processors to determine the lens position for the secondary camera lens based on the first lens position based on the comparison and the confidence value being greater than a threshold value.

25. The computer-readable storage medium of claim 20, further comprising instructions that cause the one or more processors to: determine a first confidence value associated with a first depth estimation algorithm for the secondary camera and a second confidence value associated with a second depth estimation algorithm for the secondary camera, wherein the instructions that cause the one or more processors to determine the lens position for the secondary camera lens comprise

instructions that cause the one or more processors to determine the lens position for the secondary camera lens based on the first lens position based on the comparison, the first confidence value being less than a first threshold value, and the second confidence value being greater than a second threshold value.

26. A device for camera processing, the device comprising: means for detecting a first region of interest for image content captured with a primary camera; means for detecting a second region of interest for image content captured with a secondary camera; means for comparing the first region of interest and the second region of interest; means for determining a first lens position for a secondary camera lens of the secondary camera based on an autofocus process for the secondary camera; means for determining a second lens position for the secondary camera lens of the secondary camera based on a determined lens position for a primary camera lens of the primary camera; means for determining a lens position for the secondary camera lens based on the first lens position or the second lens position based at least in part on the comparison of the first region of interest and the second region of interest; and means for causing the secondary camera to capture images with the secondary camera lens at the determined lens position.

27. The device of claim 26, further comprising: means for generating a final image having a bokeh effect based on a first image captured with the primary camera and a second image captured with the secondary camera.

28. The device of claim 26, wherein the means for comparing the first region of interest and the second region of interest comprises means for comparing an object in the first region of interest and an object in the second region of interest, and wherein the means for determining the lens position comprises means for determining the lens position for the secondary camera lens based on the first lens position based at least in part on the object in the first region of interest being the same as the object in the second region of interest.

29. The device of claim 26, wherein the means for comparing the first region of interest and the second region of interest comprises means for comparing a location of the first region of interest and a location of the second region of interest, and wherein the means for determining the lens position comprises means for determining the lens position for the secondary camera lens based on the first lens position based at least in part on the location of the first region of interest being the same as the location of the second region of interest.

30. The device of claim 26, further comprising: means for determining a confidence value associated with a depth estimation algorithm for the secondary camera, wherein the means for determining the lens position for the secondary camera lens comprises means for determining the lens position for the secondary camera lens based on the first lens position based on the comparison and the confidence value being greater than a threshold value.
